

Further Reading

The Computerphile channel on YouTube has videos on quicksort at https://youtu.be/XE4VP_8Y0BU and merge sort at https://youtu.be/kgBjXUE_Nwc. If you want a more comprehensive tutorial, the free “Algorithmic Toolbox” online course covers many of the same topics that a freshman data structures and algorithms course would cover, including binary search, quicksort, and merge sort. You can sign up for this Coursera course at <https://www.coursera.org/learn/algorithmic-toolbox>.

Sorting algorithms are often compared to each other in lessons on big O algorithm analysis, which you can read about in Chapter 13 of my book *Beyond the Basic Stuff with Python* (No Starch Press, 2020). You can read this chapter online at <https://inventwithpython.com/beyond>. Python developer Ned Batchelder describes big O and “how code slows as your data grows” in his 2018 PyCon talk of the same name at <https://youtu.be/duvZ-2UK0fc>.

Divide-and-conquer algorithms are useful because they often can be run on multiple computers in parallel. Guy Steele Jr. gives a Google TechTalk titled “Four Solutions to a Trivial Problem” on this topic at <https://youtu.be/ftcIcn8AmSY>.

Professor Tim Roughgarden produced a video lecture for Stanford University on Karatsuba multiplication at <https://youtu.be/JCbZayFr9RE>.

To help your understanding of quicksort and merge sort, obtain a pack of playing cards or simply write numbers on index cards and practice sorting them by hand according to the rules of these two algorithms. This offline approach can help you remember the pivot-and-partition of quicksort and the divide-merge of merge sort.